FEU Institute of Technology
COLLEGE OF ENGINEERING • COLLEGE OF COMPUTER STUDIES

Computer System and Architecture

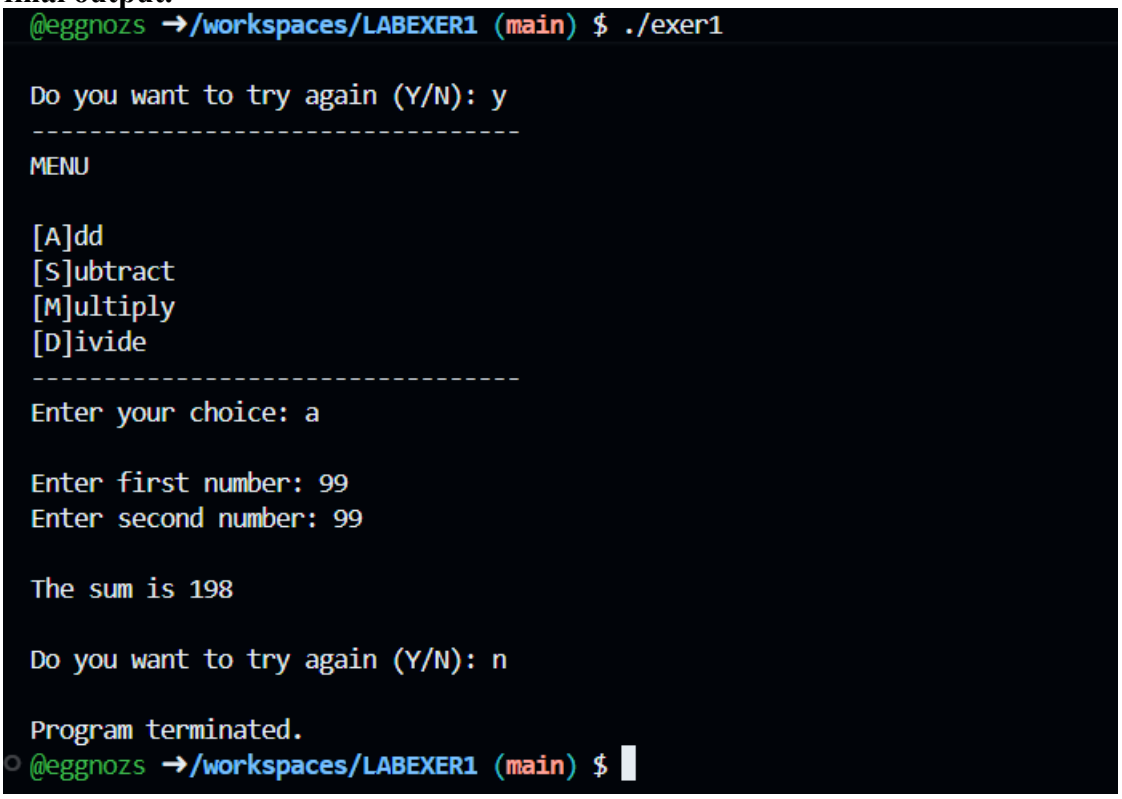
Machine Problem

MIDTERM

Name of Student	Name of Professor
Data Performed	Date Submitted

List the steps that you perform to come up with the program.

Paste the source code and explain the functionalities and take a screenshot of the final output.



```
@eggnozs →/workspaces/LABEXER1 (main) $ ./exer1
---------------------------------
MENU

[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: a

Enter first number: 12
Enter second number: 34

The sum is 46

Do you want to try again (Y/N): y
---------------------------------
MENU

[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice:
```

```
---------------------------------
MENU

[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: a

Enter first number: 2
Enter second number: 3

The sum is 5

Do you want to try again (Y/N): n

Program terminated.
@eggnozs →/workspaces/LABEXER1 (main) $ ./exer1
```

SUBTRACTION

```
---------------------------------
MENU

[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: s

Enter first number: 75
Enter second number: 28

The difference is 47

@eggnozs ➜/workspaces/LABEXER1 (main) $ ./exer1
---------------------------------
MENU

[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: s

Enter first number: 9
Enter second number: 4

The difference is 5
```

```
---------------------------------
MENU

[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: s

Enter first number: 12
Enter second number: 34

The difference is -22
```

MULTIPLICATION

```
---------------------------------
MENU

[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: m

Enter first number: 12
Enter second number: 11

The product is 132
```

```
---------------------------------
MENU

[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: m

Enter first number: 99
Enter second number: 99

The product is 9801
@eggnozs →/workspaces/LABEXER1 (main) $ ./exer1
---------------------------------
MENU

[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: m

Enter first number: 2
Enter second number: 3

The product is 6
```

DIVISION

```
@eggnozs →/workspaces/LABEXER1 (main) $ ./exer1
---------------------------------
MENU

[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: d

Enter first number: 8
Enter second number: 2

The quotient is 4 with remainder 0
```

```
---------------------------------
MENU

[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: d

Enter first number: 12
Enter second number: 0

Cannot divide by zero.
```

```
---------------------------------
MENU

[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: d

Enter first number: 25
Enter second number: 4

The quotient is 6 with remainder 1
```

The program displays a menu where the user chooses addition, subtraction, multiplication, or division. It then asks for two numbers from **0 to 99**, allowing two-digit input.

Each operation uses a separate **procedure**. The `CALL` instruction runs the selected procedure, while `RET` returns control to the main program.

- **Addition:** Uses `ADD` to calculate the sum.
- **Subtraction:** Uses `SUB` and supports negative results.
- **Multiplication:** Uses `IMUL` to calculate the product.
- **Division:** Uses `DIV` to calculate the quotient and remainder. It checks for division by zero first.

The `read_number` procedure converts typed digits into numbers. The `print_number` procedure converts the result into characters for display.

After showing the result, the program asks whether to try again. Entering **Y** clears the screen and returns to the menu. Entering **N** terminates the program. Invalid entries display an error and prompt the user again.